

Candice Xia

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EDUCATION

Master's in Mechanical Engineering

University of California, Berkeley

Concentration in Product Design | The Fung Institute for Engineering Leadership Program

Bachelor's in mechanical engineering, Minor in Management

Purdue University

Dean's List and Semester Honors; Women in Engineering Program

RELEVANT EXPERIENCES

Graduate Researcher

2025-2026

University of California, Berkeley, CA

- Open Hardware for Robotic Trash Sorting: Next Generation Smart Suction Cup
- Designed and iterated chambered suction cup prototypes for robotic trash sorting through Onshape CAD and molding-based fabrication, supporting pickup of irregularly shaped objects.
- Programmed an RP2040-based control system for the pressure-sensing subsystem and integrated the sensing hardware with a ROS2-controlled robotic arm.

Mechanical R&D Engineer & Head Research Intern

2020-2024

Zonlink Gear Wheel Transmission, Chongqing

- Designed and machined components, cutting tools, and inspection fixtures for internal gear manufacturing, implementing process improvements that reduced setup time by 30% and improved workpiece stability by 15%.
- Led design iteration and process optimization efforts in collaboration with senior engineers, while mentoring 5 junior interns to improve manufacturing accuracy and efficiency.
- Contributed to 4 utility patents related to gear manufacturing and tooling design (CN212350651U / CN212363066U / CN111195736B / CN111842522A).

Undergraduate Researcher

Mechatronic and Fluid Dynamic Investigation of a Battery-Powered Air Purification and Active Fragrance Delivery Unit

Purdue University, IN

- Developed a dual-function air purification and fragrance delivery system, including mechanical design of a motorized mixing mechanism for active dispersion of zeolite-based scent carriers.
- Ran CFD simulations with a Moving Reference Frame model to improve airflow performance and used injection molding analysis to evaluate part manufacturability.

OTHER RELATED PROJECTS

Hands-Free Assistive Walking Device for Knee Rehabilitation

- Performed mechanical design and calculations for a knee rehabilitation assistive device, contributing to an adjustable spring-linkage structure designed to offload the injured knee during walking.
- Iterated CAD models and 3D-printed components across multiple prototype cycles to improve fit, structural performance, and overall device functionality.

Embedded Haptic Surgical Skills Trainer

- Developed a real-time surgical training prototype that quantified grip force and hand steadiness during laparoscopic tasks, enabling objective performance feedback through 100 Hz sensor streaming and immediate haptic alerts.
- Built an ESP32/FreeRTOS and Python system that fused force and motion sensing into a live training dashboard, reducing reliance on subjective instructor feedback while preserving realistic tool use by leaving the surgical instrument unmodified.

SKILLS

- **CAD:** OnShape, Siemens NX, SOLIDWORKS, Fusion360, PTC Creo
- **Engineering Tools:** KiCAD PCB, ABAQUS FEA, Python, MATLAB
- **Hardware:** ESP32, RP2040, Arduino, Soldering, 3D Printing
- **Languages:** English, Mandarin Chinese (Bilingual)